

SNP Filters: Identity Analyses

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Filtering the VCF

The raw VCF file is quite large (over 8Gb). We want to subset this large dataset to a set of markers that will be maximally informative for identity analyses.

The basic steps involved in the filters below include:

0. Cleaning the raw VCF - remove duplicated lines, half-calls, indels
1. Quality filter at the locus level - drop the SNP locus if quality < 30
2. Maximum depth filter at the locus level - drop SNP loci if total sequencing depth is $> 2\sigma$ above mean total depth
3. Min depth filter at the genotype level - don't call genotypes with < 10 reads
4. Missing data filter - drop SNPs missing genotypes for $> 30\%$ of samples
5. Polymorphism filter - only retain loci that have exactly two alleles
6. Minor allele frequency filter - keep sites with MAF > 0.2
7. Avoiding linkage - prune markers based on LD within 500kb windows ($r^2 < 0.8$)

After all of these filters, the file `LGC_SCR.vcf` will contain SNP genotypes for samples processed by LGC / Rapid Genomics. In the next step, we will combine these with a smaller number of loci genotyped via GT-seq (by GTSeek). We will come back to the raw VCF used here to extract genotypes for loci from the GT-seq panel that did not pass the filters used in creating `LGC_SCR.vcf` (due to minor allele frequencies, missing data levels, etc.). We will also apply individual-level filters for missingness and excessively high heterozygosity in the next step after creating the full, combined vcf file containing all samples and loci.

Step 0: Clean the raw VCF

First, copy the files and create some summaries of average depth, per-individual missing data, and per-genotype depth.

```
cd /scratch/jdrobins/CGA_SCR/snpfiltering

cp ../../LGC_BlackBears/UGA_173101/Delivery-20250620_merged/VCFs/UGA_173101_SNPs_Raw.vcf ./

#Use bcftools norm to remove duplicate SNPs
bcftools norm -d snps UGA_173101_SNPs_Raw.vcf -o filt0uniq.vcf

#Use sed to replace the half-calls (./0 and ./1) with missing data (./.)
grep ^# filt0uniq.vcf > filt0.vcf
grep -v ^# filt0uniq.vcf | sed -e 's/\./\.[0,1]/\./\./g' | sed -e 's/[0,1]\.\./\./g' >> filt0.vcf

#Use vcftools to remove indels
vcftools --vcf filt0.vcf --remove-indels --recode --out filt0

#make .idepth, .imiss, and .gdepth (this one is bigger) files
vcftools --vcf filt0.recode.vcf --out raw --depth
vcftools --vcf filt0.recode.vcf --out raw --missing-indv
vcftools --vcf filt0.recode.vcf --out raw --geno-depth

#How many SNPs do we start with?
grep -v ^# filt0.recode.vcf | wc -l
```

Lines total/split/joined/realigned/mismatch_removed/dup_removed/skipped: 319680/0/0/0/0/5871/0

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```
--vcf filt0.vcf
--out filt0
--recode
--remove-indels
```

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen

```

Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 312565 out of a possible 313809 Sites
Run Time = 588.00 seconds

```

VCFTools - 0.1.17

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Parameters as interpreted:

```

--vcf filt0.recode.vcf
--depth
--out raw

```

```

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting Mean Depth by Individual
After filtering, kept 312565 out of a possible 312565 Sites
Run Time = 26.00 seconds

```

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```

--vcf filt0.recode.vcf
--missing-indv

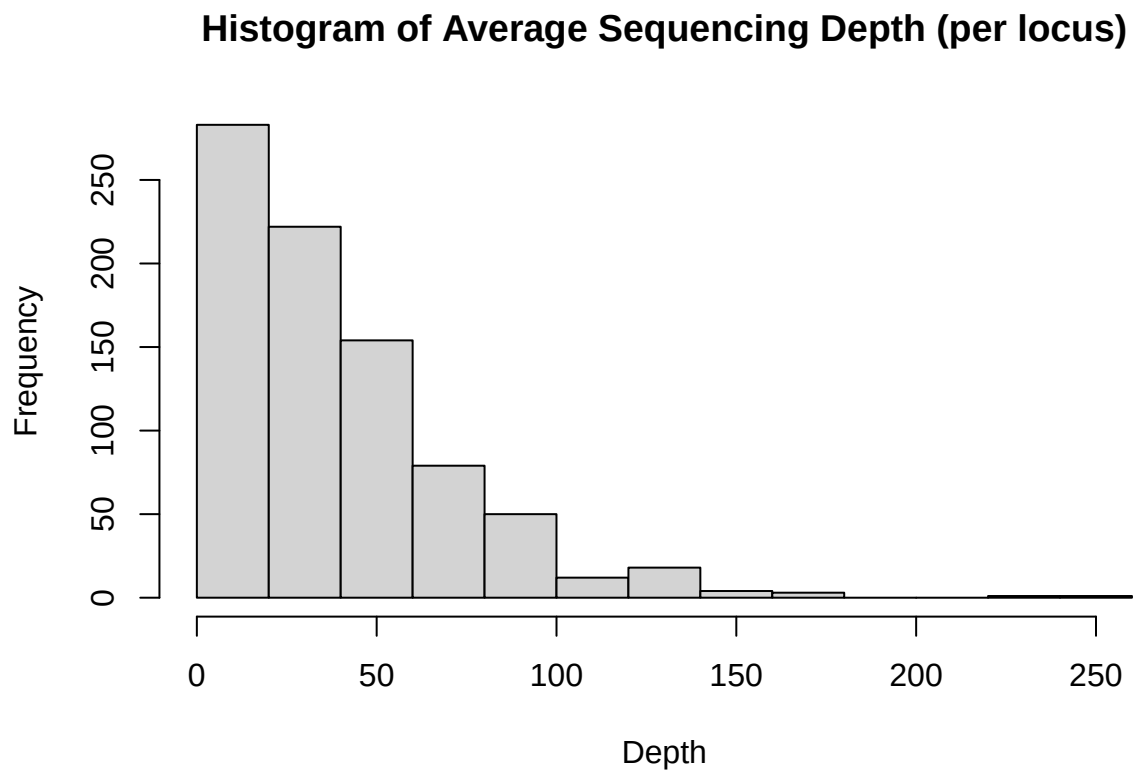
```


Outputting Depth for Each Genotype
After filtering, kept 312565 out of a possible 312565 Sites
Run Time = 50.00 seconds
312565

Plot some summaries of depth and missingness...

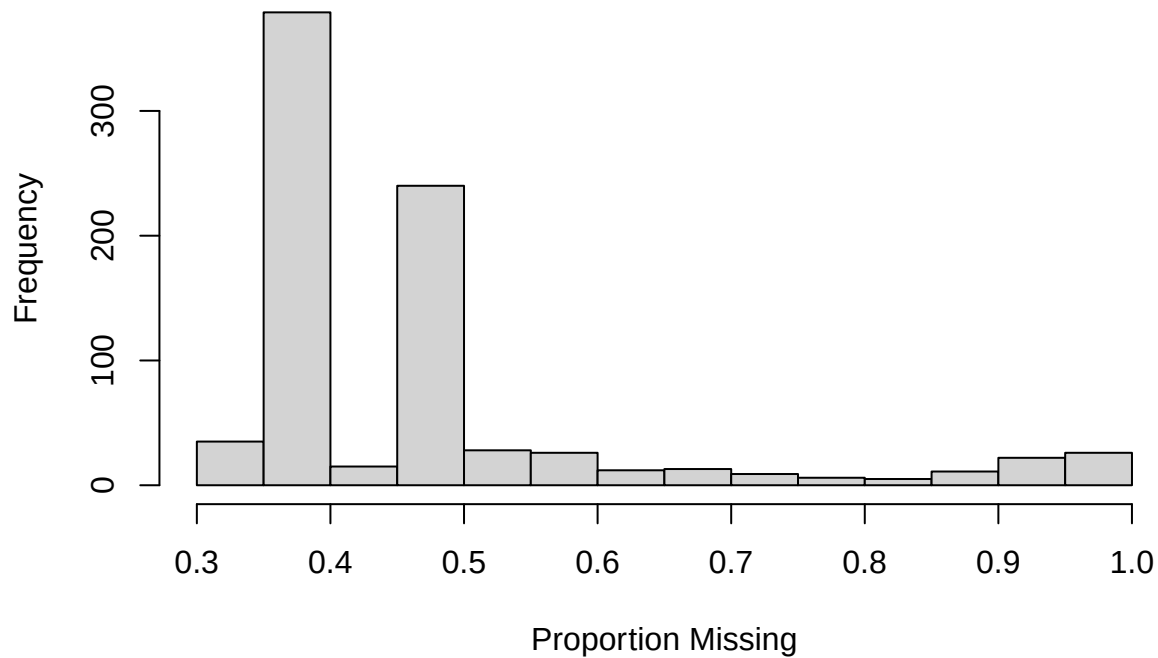
```
depth <- read.table("raw.iddepth", header = TRUE)
miss <- read.table("raw.imiss", header = TRUE)

hist(depth$MEAN_DEPTH, main = "Histogram of Average Sequencing Depth (per locus)",
      xlab = "Depth")
```



```
hist(miss$F_MISS, main = "Histogram of Missing Data (per individual)",
      xlab = "Proportion Missing")
```

Histogram of Missing Data (per individual)



```
#Quantiles of depth and missingness
quantile(depth$MEAN_DEPTH)
```

```
##          0%          25%          50%          75%          100%
##  1.70952  14.48280  31.35840  52.97025  255.97600
```

```
quantile(miss$F_MISS)
```

```
##          0%          25%          50%          75%          100%
##  0.3491590  0.3525955  0.3999840  0.4762450  0.9961320
```

Step 1: Genotype Quality

The first filter we apply is a genotype quality filter (`--minQ`).

Prior to these filters, we have 3.12565×10^5 loci in the VCF...

```
#Filter sites
# min site quality of 30
vcftools --vcf filt0.recode.vcf --out filt1 \
  --minQ 30 --remove-filtered-all --recode
```

```
VCftools - 0.1.17
```

```
Parameters as interpreted:
--vcf filt0.recode.vcf
--minQ 30
--out filt1
--recode
--remove-filtered-all
```

Step 2: Maximum Depth

Prior to these filters, we have 9.6917×10^4 SNPs in the dataset...

VCFtools - 0.1.17

```
Parameters as interpreted:
  --vcf filt1.recode.vcf
  --out filt1
  --site-depth
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting Depth for Each Site
After filtering, kept 96917 out of a possible 96917 Sites
Run Time = 8.00 seconds
```

```
ldep <- read.table("filt1.ldepth", header = TRUE)
```

```
## [1] 20672.05
```

```
#This will be our threshold...
# nothing more than 2 sd above the mean...
cap <- mean(ldep$SUM_DEPTH) + (2*sd(ldep$SUM_DEPTH))
cap
```

```
keeper.sites <- ldep[which(ldep$SUM_DEPTH <= cap),c(1,2)]
write.table(keeper.sites, file="keepersites.txt",
            quote=FALSE, row.names = FALSE, col.names = FALSE)
```



```
#Then remove loci based on this threshold...
# only keep the SNPs in the list created above
vcftools --vcf filt1.recode.vcf --out filt2 \
--positions keepsites.txt \
--remove-filtered-all --recode
```

VCFtools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```
--vcf filt1.recode.vcf
--out filt2
--positions keepersites.txt
--recode
--remove-filtered-all
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 92692 out of a possible 96917 Sites
Run Time = 177.00 seconds
```

Step 3: Genotype Depth

In the next filter, we remove genotype calls that are based on fewer than 10 reads. We set this threshold to maximize our confidence in individual calls. This is very important for identity analyses where genotyping errors lead to mismatches and may inflate the number of unique individuals detected in our project.

Prior to these filters, we have 9.2692×10^4 loci in the VCF...

```
#Min depth for genotype calls is 10 reads
vcftools --vcf filt2.recode.vcf --out filt3 \
    --minDP 10 --remove-filtered-all --recode
```

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```
--vcf filt2.recode.vcf
--minDP 10
--out filt3
--recode
--remove-filtered-all
```

Step 4: Missing Data (per locus)

Prior to these filters, we have 9.2692×10^4 loci in the VCF...

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```
--vcf filt4.recode.vcf
--freq
--out filt4
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting Frequency Statistics...
After filtering, kept 18516 out of a possible 18516 Sites
Run Time = 1.00 seconds
```

Now read this file into R and identify sites that have exactly two alleles in the dataset...

```
snpfreq <- read.table("filt4.frq", skip = 1, fill=TRUE)
colnames(snpfreq) <- c("CHROM","POS","N_ALLELES","N_CHR", "A1","A2","A3")
biallelic <- snpfreq[which(snpfreq$N_ALLELES == 2),c(1,2)]

fixed.sites <- c(grep(":1", biallelic$A1),
                 grep(":1", biallelic$A2),
                 grep(":1", biallelic$A3))
fixed.sites
```

```
## integer(0)
```

```
biallelic[fixed.sites,]
```

```
## [1] CHROM POS
## <0 rows> (or 0-length row.names)
```

```
if(length(fixed.sites) > 0) {
  biallelic <- biallelic[-fixed.sites,c(1,2)]
}

nrow(biallelic)
```

```
## [1] 13890
```

```
write.table(biallelic, file = "biallelic.txt",
           quote = FALSE, row.names = FALSE, col.names = FALSE)
```

Now retain only SNP loci in this list ...

```
#Only retain SNPs in the file created above
vcftools --vcf filt4.recode.vcf --out filt5 \
  --positions biallelic.txt \
  --remove-filtered-all --recode
```

VCFtools - 0.1.17

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Parameters as interpreted:

```
--vcf filt4.recode.vcf
--out filt5
--positions biallelic.txt
--recode
--remove-filtered-all
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 13890 out of a possible 18516 Sites
Run Time = 29.00 seconds
```

Step 6: Minor Allele Frequency

The power for individual identification is maximized in biallelic loci with equal allele frequencies ($p = q = 0.5$). Thus, loci with high minor allele frequencies (MAF) will be the most useful for identifying recaptures. Here, we remove sites with $MAF < 0.2$. Note that the GT-seq panel that we're using has some low frequency SNPs included. In subsequent steps (*i.e.*, `4_combineGTSeek`), we'll come back to the raw VCF and extract genotypes for markers that are excluded here to retain SNPs that were genotyped via GT-seq. However, the bulk of our dataset will have high MAF, maximizing the information content of each genotyped SNP locus.

Prior to these filters, we have 1.389×10^4 loci in the VCF...

```
#MAF filter - remove SNPs with minor allele frequency < 0.2
vcftools --vcf filt5.recode.vcf --out filt6 \
        --maf 0.2 --remove-filtered-all --recode
```

VCftools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```
--vcf filt5.recode.vcf
--maf 0.2
--out filt6
--recode
--remove-filtered-all
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 986 out of a possible 13890 Sites
Run Time = 3.00 seconds
```

Step 7: Pruning based on Linkage Disequilibrium (LD)

Finally, tightly linked loci do not provide independent information on individual identity, as they are inherited together. Thus, we need to subsample the SNP loci to avoid markers that are tightly linked. Below, we use PLINK2 to prune markers that exhibit high LD within 500 kb windows ($r^2 < 0.8$).

Prior to these filters, we have 986 loci in the VCF...

```
#Create pruning files (LDprune prefix) with sets of markers
# to remove (prune.out suffix) and keep (prune.in suffix)
plink2 --vcf filt6.recode.vcf --allow-extra-chr \
        --indep-pairwise 500 kb 1 0.8 \
```

```

--set-all-var-ids @:# --out LDprune --silent

#Apply the prune.in file to keep only loci that pass the LD filter
plink2 --vcf filt6.recode.vcf --set-all-var-ids @:# \
--allow-extra-chr --extract LDprune.prune.in \
--recode vcf-4.2 --out LGC_SCR --silent

#Extract allele frequencies for these SNPs to calculate p_{ID}
plink2 --vcf LGC_SCR.vcf --allow-extra-chr --freq --out LGC_SCR --silent

#How many loci?
grep -v ^# LGC_SCR.vcf | wc -l

```

831

After all filters are applied, we have 831 loci in the VCF. These will be combined with SNPs genotyped via GT-seq to form the dataset we use for the recapture analyses.

Summarize Dataset

Now we'll i) summarize the extent of missing data in our filtered VCF and ii) record the number of genotyped markers for each sample in our metadata file.

Missingness

After combining SNP genotypes from LGC / RapidGenomics with those from GTSeek, we will impose a filter to drop samples based on missing data. As a preliminary assessment of the number of samples we're likely to lose, we now evaluate the proportion of missing data in the LGC SNP set.

First, create files with summaries of the missing data levels by locus and by sample.

```

#Missing data by locus
vcftools --vcf LGC_SCR.vcf --missing-site --out LGC_SCR

#Missing data by sample
vcftools --vcf LGC_SCR.vcf --missing-indv --out LGC_SCR

```

VCFtools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```

--vcf LGC_SCR.vcf
--out LGC_SCR
--missing-site

```

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allele frequency in the sample (scaled by max per sample number of alleles)"

Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allele proportion"


```

Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
After filtering, kept 827 out of 827 Individuals
Outputting Site Missingness
After filtering, kept 831 out of a possible 831 Sites
Run Time = 0.00 seconds

```

VCfTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```

--vcf LGC_SCR.vcf
--missing-indv
--out LGC_SCR

```

```

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
After filtering, kept 827 out of 827 Individuals
Outputting Individual Missingness
After filtering, kept 831 out of a possible 831 Sites
Run Time = 0.00 seconds

```

By locus

Read this into R and check the mean and range of the proportion of genotypes missing across loci.

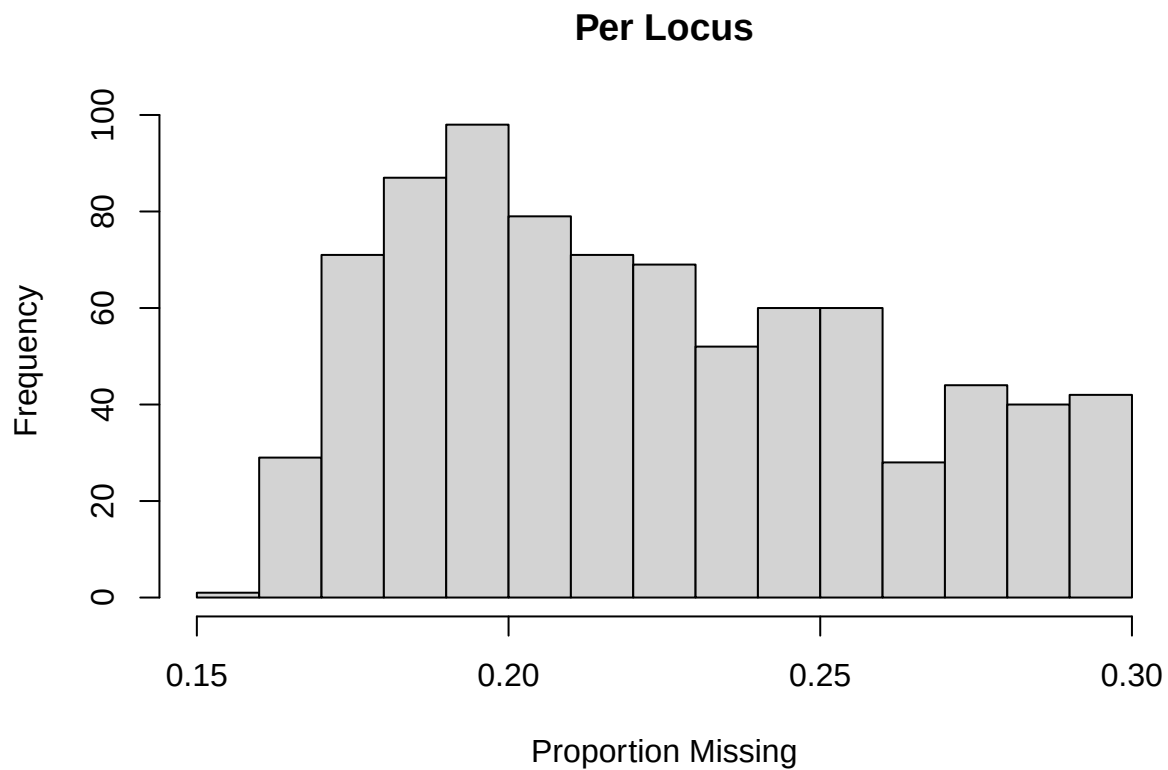

```
missLocus <- read.table("LGC_SCR.lmiss", header = TRUE)
mean(missLocus$F_MISS)
```

```
## [1] 0.2228474
```

```
range(missLocus$F_MISS)
```

```
## [1] 0.155985 0.299879
```

```
hist(missLocus$F_MISS,
     xlab = "Proportion Missing", ylab = "Frequency", main = "Per Locus")
```



By sample

Now the proportion of missing data for each sample.

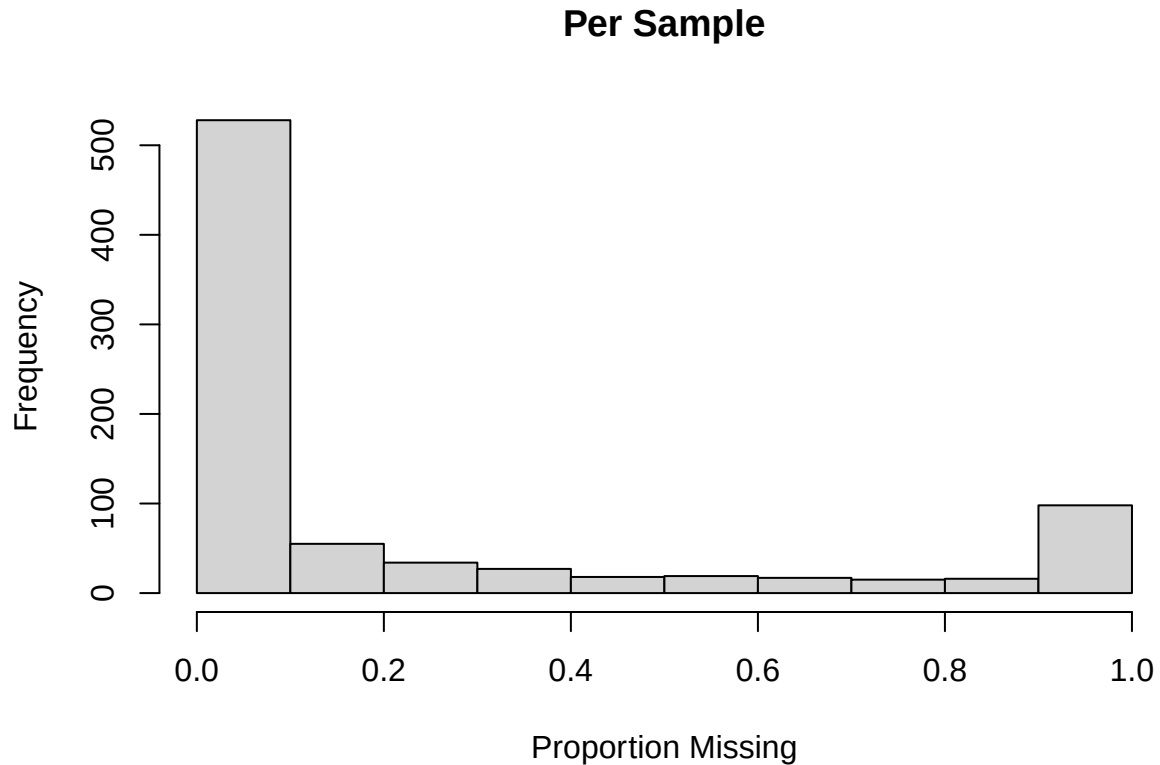
```
missSamp <- read.table("LGC_SCR.imiss", header = TRUE)
mean(missSamp$F_MISS)
```

```
## [1] 0.2228474
```

```
range(missSamp$F_MISS)
```

```
## [1] 0 1
```

```
hist(missSamp$F_MISS,  
     xlab = "Proportion Missing", ylab = "Frequency", main = "Per Sample")
```



```
#How many samples do we lose with various missing data filters...  
nrow(missSamp) #total samples in the VCF
```

```
## [1] 827
```

```
#Lost if keeping samples...  
sum(missSamp$F_MISS >= 0.9) # with <90% missing
```

```
## [1] 98
```

```
sum(missSamp$F_MISS >= 0.7) # with <70% missing
```

```
## [1] 129
```

```
sum(missSamp$F_MISS >= 0.5) # with <50% missing
```

```
## [1] 165
```

```
sum(missSamp$F_MISS >= 0.3) # with <30% missing
```

```
## [1] 210
```

```
sum(missSamp$F_MISS >= 0.1) # with <10% missing
```

```
## [1] 299
```

Given that the genotypes we are working with come from non-invasive (hair) samples, higher thresholds for successful genotyping are unrealistic. We will apply an intermediate threshold in the next step of the filtering pipeline (after incorporating samples genotyped by GTSeek).

Record metadata

Finally, add information on the number of SNPs genotyped in each sample to our sample metadata file.

```
smd <- read.csv("../sampleinfo/sampleMetadata_basic.csv", header = TRUE)

#Add the number of genotyped SNPs in the filtered VCF to sampleMetaData
smd$nSNPgt <- rep(NA, nrow(smd))
for(i in 1:nrow(smd)) {
  tmp <- which(missSamp$INDV == smd$rgID[i])
  smd$nSNPgt[i] <- missSamp$N_DATA[tmp]-missSamp$N_MISS[tmp]
  rm(tmp)
}
write.csv(smd, "sampleMetadata_nSNPgt.csv", quote = FALSE, row.names=FALSE)
```